

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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25ITR138

JAVA ASSIGNMENT

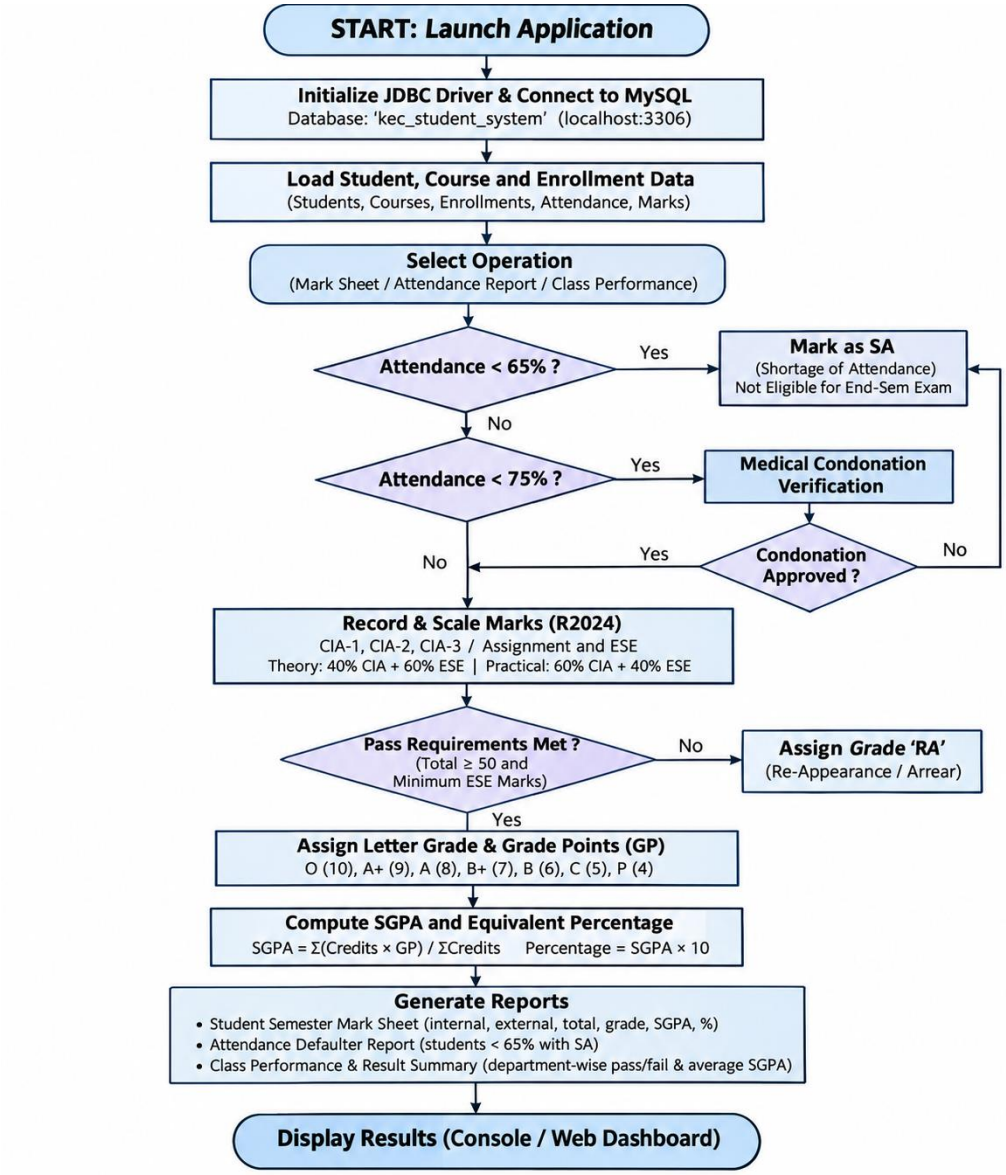
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- ❖ Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.
- ❖ Establish a connection to the kec_student_system database running on the local MySQL server using DriverManager.getConnection().
- ❖ Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, course title, credits, and course type.
- ❖ Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.
- ❖ Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).
- ❖ Record CIA-1, CIA-2, CIA-3, assignment marks, and end-semester examination (ESE) marks. Calculate scaled internal assessment marks according to the specified theory and practical course weightages.
- ❖ Verify the minimum passing requirements for the end-semester examination and scale the marks according to the applicable theory and practical regulations.
- ❖ Calculate total marks by combining scaled internal assessment and ESE marks. Assign letter grades and grade points according to the specified grading scale, and mark students as RA if they fail the examination or score below the required passing mark.
- ❖ Generate student semester mark sheets containing internal marks, external marks, total marks, grades, pass/fail status, SGPA, and equivalent percentage.
- ❖ Generate attendance defaulter reports listing students below 65% attendance and flagging their SA status.
- ❖ Generate class performance and result summary reports containing department-wise pass/fail counts and class average SGPA.

DATABASE SCHEMA:

1.STUDENTS TABLE:

Your SQL query has been executed successfully.

```
DESCRIBE students;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
roll_number	varchar(20)	NO	PRI	*NULL*	
name	varchar(100)	NO		*NULL*	
department	varchar(10)	NO		*NULL*	
current_semester	int(11)	NO		*NULL*	

2. COURSES TABLE

Your SQL query has been executed successfully.

```
DESCRIBE courses;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
course_code	varchar(20)	NO	PRI	*NULL*	
title	varchar(200)	NO		*NULL*	
credits	int(11)	NO		*NULL*	
course_type	enum('THEORY','PRACTICAL','EMBEDDED')	NO		*NULL*	

3. ENROLLMENTS TABLE:

```
DESCRIBE enrollments;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
enrollment_id	int(11)	NO	PRI	*NULL*	auto_increment
roll_number	varchar(20)	NO	MUL	*NULL*	
course_code	varchar(20)	NO	MUL	*NULL*	
semester	int(11)	NO		*NULL*	
total_classes	int(11)	NO		0	
attended_classes	int(11)	NO		0	

4. ASSESSMENT MARKS TABLE:

Your SQL query has been executed successfully.

```
DESCRIBE assessment_marks;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
enrollment_id	int(11)	NO	PRI	*NULL*	
phase	enum('CIA1','CIA2','CIA3','ESE')	NO	PRI	*NULL*	
marks	int(11)	YES		*NULL*	

SOURCE CODE:

1.)MainApp.java:

```
 package com.kec;

import com.kec.dao.*;

import com.kec.model.*;
import java.util.*;

public class MainApp {
    private static final Scanner = new Scanner(System.in);
    private static final StudentDAO = new StudentDAO();
    private static final CourseDAO = new CourseDAO();
    private static final AcademicDAO = new AcademicDAO();
    public static void main(String[] args) {
        System.out.println("\nKONGU ENGINEERING COLLEGE (AUTONOMOUS), PERUNDURAI");
        System.out.println("Student Academic & Continuous Evaluation System (R2024)");
        while (true) {
            printMenu();
            switch (readInt("Enter your choice: ")) {
                case 1 -> addStudent();
                case 2 -> addCourse();
                case 3 -> enrollStudent();
                case 4 -> updateAttendance();
                case 5 -> enterAllMarks();
                case 6 -> generateMarkSheet();
                case 7 -> attendanceDefaulterReport();
                case 8 -> classPerformanceReport();
                case 9 -> viewAllStudents();
                case 10 -> viewAllCourses();
                case 11 -> listEnrollments();
                case 0 -> { System.out.println("Goodbye!"); return; }
                default -> System.out.println("Invalid choice. Try again.");
            }
        }
    }
```

```
 private static void printMenu() {
      System.out.println("""
private static void addStudent() {
      String roll = readLine("Roll Number: ").toUpperCase();
      String name = readLine("Student Name: ");
      String dept = readLine("Department: ").toUpperCase();

     int sem = readInt("Current Semester: ");
      Student = new Student(roll, name, dept, sem);
      if (studentDAO.addStudent(student))
         System.out.println("Student added successfully.");
  }
  private static void addCourse() {
      String code = readLine("Course Code: ").toUpperCase();
      String title = readLine("Course Title: ");
      int credits = readInt("Credits: ");
      String type = readLine("Course Type (THEORY/PRACTICAL/EMBEDDED): ").toUpperCase();
      Course = new Course(code, title, credits, type);
      if (courseDAO.addCourse(course))
         System.out.println("Course added successfully.");
  }
  private static void enrollStudent() {
      String roll = readLine("Student Roll Number: ");
      String code = readLine("Course Code: ");
      int sem = readInt("Semester: ");
      if (academicDAO.enrollStudent(roll, code, sem))
         System.out.println("Student enrolled successfully.");
  }
  private static void updateAttendance() {
      String roll = readLine("Student Roll Number: ");
      int sem = readInt("Semester: ");
      academicDAO.listEnrollments(roll, sem);
      int id = readInt("Enrollment ID: ");
      int total = readInt("Total Classes Held: ");
      int attended = readInt("Classes Attended: ");
      if (total < 0 || attended < 0 || attended > total) {
         System.out.println("Invalid attendance values.");
         return;
      }
      if (academicDAO.updateAttendance(id, total, attended)) {
         double pct = total == 0 ? 0 : attended * 100.0 / total;
         System.out.printf("Attendance: %.1f%%%n", pct);
         if (pct < 65) System.out.println("Warning: Shortage of Attendance (SA).");
         else if (pct < 75) System.out.println("Medical condonation verification may be required.");
      }
  }
 private static void enterAllMarks() {
      String roll = readLine("Student Roll Number: ");
      int sem = readInt("Semester: ");
      academicDAO.listEnrollments(roll, sem);
      int id = readInt("Enrollment ID: ");
      academicDAO.showCurrentMarks(id);
      int cia1 = readMarks("CIA-1 Marks (0-100): ");
      int cia2 = readMarks("CIA-2 Marks (0-100): ");
      int cia3 = readMarks("Assignment/Quiz Marks (0-100): ");
      int ese = readMarks("ESE Marks (0-100): ");
      boolean ok = academicDAO.recordMarks(id, "CIA1", cia1);
      ok &= academicDAO.recordMarks(id, "CIA2", cia2);
      ok &= academicDAO.recordMarks(id, "CIA3", cia3);

      ok &= academicDAO.recordMarks(id, "ESE", ese);
      System.out.println(ok ? "All marks recorded." : "Some marks could not be saved.");
  }
```

```
    private static int readMarks(String prompt) {
        while (true) {
            int mark = readInt(prompt);
            if (mark >= 0 && mark <= 100) return mark;
            System.out.println("Enter a mark between 0 and 100.");
        }
    }
    private static void generateMarkSheet() {
        String roll = readLine("Student Roll Number: ");

         int sem = readInt("Semester: ");
        academicDAO.generateMarkSheet(roll, sem);
    }
    private static void attendanceDefaulterReport() {
        academicDAO.attendanceDefaulterReport(readInt("Semester: "));
    }
    private static void classPerformanceReport() {
        academicDAO.classPerformanceReport(readInt("Semester: "));
    }
    private static void viewAllStudents() {
        List<Student> students = studentDAO.getAllStudents();
        System.out.printf("%-15s %-25s %-10s %-5s%n", "Roll No", "Name", "Dept", "Sem");
        for (Student s : students)
            System.out.printf("%-15s %-25s %-10s %-5d%n",
                s.getRollNumber(), s.getName(), s.getDepartment(), s.getCurrentSemester());
        System.out.println("Total students: " + students.size());
    }
    private static void viewAllCourses() {
        List<Course> courses = courseDAO.getAllCourses();
        System.out.printf("%-12s %-35s %-7s %-12s%n", "Code", "Title", "Credits", "Type");
        for (Course c : courses)
            System.out.printf("%-12s %-35s %-7d %-12s%n",
                c.getCourseCode(), c.getTitle(), c.getCredits(), c.getCourseType());
        System.out.println("Total courses: " + courses.size());
    }
    private static void listEnrollments() {
        String roll = readLine("Student Roll Number: ");
        int sem = readInt("Semester: ");
        academicDAO.listEnrollments(roll, sem);
    }
    private static String readLine(String prompt) {
        System.out.print(prompt);
        return scanner.nextLine().trim();
    }
    private static int readInt(String prompt) {
        while (true) {
            try {
                System.out.print(prompt);
                return Integer.parseInt(scanner.nextLine().trim());
            } catch (NumberFormatException e) {
                System.out.println("Please enter a valid integer.");
            }
        }
    }
}
```

2.)DBConnection.java

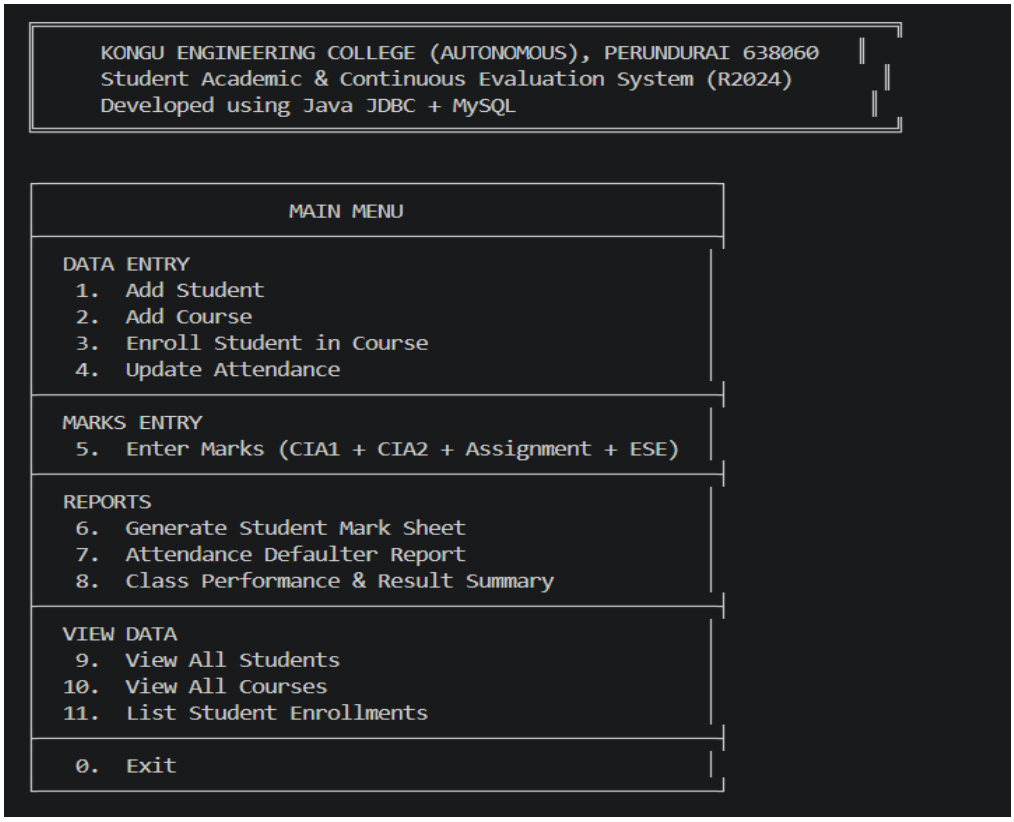
```
package com.kec;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
public class DBConnection {


    private static final String DB_URL =
"jdbc:mysql://localhost:3306/kec_student_system?useSSL=false&allowPublicKeyRetrieval=true";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "";

static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("MySQL JDBC Driver not found. Ensure the connector is on the
classpath.");
            e.printStackTrace();
        }
    }
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(DB_URL, DB_USER, DB_PASSWORD);
    }
```

OUTPUT SCREENSHOTS:

MAIN MENU:



STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
Enter your choice: 1

--- Add New Student ---
Roll Number (e.g., 24ITR005): 24ITR005
Student Name: Vivek Shankar S
Department (IT/CSE/ECE/MECH): IT
Current Semester: 3
[SUCCESS] Student 'Vivek Shankar S' (24ITR005) added successfully.
```

2.ADDING COURSE:

```
Enter your choice: 2

--- Add New Course ---
Course Code (e.g., 24ITR33): 24ITR33
Course Title: Artficial Intelligence
Credits: 3
Course Type (THEORY/PRACTICAL/EMBEDDED): THEORY
[SUCCESS] Course 'Artficial Intelligence' (24ITR33) added successfully.
```

3.COURSE ENROLLMENT :

```
Enter your choice: 3

--- Enroll Student in Course ---
Student Roll Number: 24ITR005
Course Code: 24ITR31
Semester: 3
[SUCCESS] Student 24ITR005 enrolled in 24ITR31 for Semester 3.
```

INCREMENTAL MARKS ENTRY: (CIA1+CIA2+Assignment+ESE)

```
Enter your choice: 5

--- Enter Marks (CIA1 + CIA2 + Assignment + ESE) ---
Student Roll Number: 24ITR005
Semester: 3
Enrollments for 24ITR005:
+------+----------+-----------------------------------+------+-----------+
| E.ID | Code     | Title                             | Cred | Type      |
+------+----------+-----------------------------------+------+-----------+
| 41   | 24ITR33  | Artficial Intelligence            | 3    | THEORY    |
+------+----------+-----------------------------------+------+-----------+
Enter Enrollment ID: 41
Currently recorded marks:
  CIA1 : 82/100
  CIA3 : 80/100
  ESE : 85/100

Enter all marks below (each out of 100):
CIA Test 1 Marks       (0-100): 82
CIA Test 2 Marks       (0-100): 80
Assignment/Quiz Marks  (0-100): 80
End Semester Exam (ESE) (0-100): 85

[SUCCESS] All marks recorded for Enrollment ID 41:
          CIA1 = 82/100  |  CIA2 = 80/100  |  Assignment = 80/100  |  ESE = 85/100
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE DEFAULTER REPORT :

```
Enter your choice: 7

--- Attendance Defaulter Report ---
Semester: 3

          ATTENDANCE DEFAULTER REPORT ? SEMESTER 3  (Attendance < 65%)
          Students flagged with SA (Shortage of Attendance) Grade

 Roll No    | Name            | Dept | Course Code | Course Title            | Att%
 24CSE003   | Harish Kumar    | CSE  | 24CSR32     | Computer Networks       |  50.0%
 24ECE003   | Ravi Shankar    | ECE  | 24ECR31     | Signals and Systems     |  56.0%
 24ECE003   | Ravi Shankar    | ECE  | 24ECR32     | Digital Electronics     |  62.0%
 24ITR003   | Karthik Raja    | IT   | 24ITR32     | Database Management Systems |  60.0%

 Total SA Entries: 4
```

2.STUDENT SEMESTER MARK SHEET :

```
Enter your choice: 6

--- Generate Student Semester Mark Sheet ---
Student Roll Number: 24ITR001
Semester: 3

            KONGU ENGINEERING COLLEGE (AUTONOMOUS), PERUNDURAI ? 638060

                STUDENT SEMESTER MARK SHEET (R2024)

 Roll No   : 24ITR001              Name       : Arun Kumar
 Department: IT                    Semester   : 3

 Code     | Title                    | Cred | Att%  | CIA  | ESE  | Total | Grade | GP | P/F

 24ITE31  | Object Oriented Programming | 3    |  95.0 | 45.0 | 45.5 |  90.5 | A+    |  9 | PASS

 24ITP31  | Data Structures Lab        | 2    |  93.3 | 55.4 | 35.2 |  90.6 | A+    |  9 | PASS
 24ITR31  | Data Structures            | 4    |  96.0 | 36.9 | 56.4 |  93.3 | O     | 10 | PASS
 24ITR32  | Database Management Systems | 4    |  92.0 | 35.1 | 54.0 |  89.1 | A+    |  9 | PASS

 Total Credits: 13     |   Passed: 4   |   Failed: 0
 SGPA: 9.31            |   Equivalent Percentage: 93.08%
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
Enter your choice: 8

--- Class Performance & Result Summary ---
Semester: 3
```

CLASS PERFORMANCE & RESULT SUMMARY ? SEMESTER 3

Department	Students	All Pass	Partial	All Fail	Avg SGPA	Avg %
CSE	3	2	1	0	7.77	77.7%
ECE	3	2	0	1	5.95	59.5%
IT	5	4	1	0	7.58	75.8%

STUDENT-WISE SGPA DETAILS

Roll No	Name	Dept	SGPA	Pct%	Status
24CSE001	Naveen Raj	CSE	10.00	100.0%	ALL CLEAR
24CSE002	Sowmiya Devi	CSE	9.00	90.0%	ALL CLEAR
24CSE003	Harish Kumar	CSE	4.31	43.1%	PARTIAL
24ECE001	Surya Prakash	ECE	10.00	100.0%	ALL CLEAR
24ECE002	Meena Kumari	ECE	7.85	78.5%	ALL CLEAR
24ECE003	Ravi Shankar	ECE	0.00	0.0%	ALL FAIL
24ITR001	Arun Kumar	IT	9.31	93.1%	ALL CLEAR
24ITR002	Priya Dharshini	IT	7.85	78.5%	ALL CLEAR
24ITR003	Karthik Raja	IT	4.92	49.2%	PARTIAL
24ITR004	Deepa Lakshmi	IT	6.85	68.5%	ALL CLEAR
24ITR005	Vivek Shankar S	IT	9.00	90.0%	ALL CLEAR

DATABASE:

STUDENTS DATABASE:

```
SELECT * from students;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	roll_number	name	department	current_semester
Edit Copy Delete	24CSE001	Naveen Raj	CSE	3
Edit Copy Delete	24CSE002	Sowmiya Devi	CSE	3
Edit Copy Delete	24CSE003	Harish Kumar	CSE	3
Edit Copy Delete	24ECE001	Surya Prakash	ECE	3
Edit Copy Delete	24ECE002	Meena Kumari	ECE	3
Edit Copy Delete	24ECE003	Ravi Shankar	ECE	3
Edit Copy Delete	24ITR001	Arun Kumar	IT	3
Edit Copy Delete	24ITR002	Priya Dharshini	IT	3
Edit Copy Delete	24ITR003	Karthik Raja	IT	3
Edit Copy Delete	24ITR004	Deepa Lakshmi	IT	3
Edit Copy Delete	24ITR005	Vivek Shankar S	IT	3

COURSES DATABASE:

```
SELECT * FROM courses;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	course_code	title	credits	course_type
Edit Copy Delete	24CSE31	Full Stack Development	3	EMBEDDED
Edit Copy Delete	24CSP31	OS Lab	2	PRACTICAL
Edit Copy Delete	24CSR31	Operating Systems	4	THEORY
Edit Copy Delete	24CSR32	Computer Networks	4	THEORY
Edit Copy Delete	24ECE31	Embedded System Design	3	EMBEDDED
Edit Copy Delete	24ECP31	Electronic Circuits Lab	2	PRACTICAL
Edit Copy Delete	24ECR31	Signals and Systems	4	THEORY
Edit Copy Delete	24ECR32	Digital Electronics	4	THEORY
Edit Copy Delete	24ITE31	Object Oriented Programming	3	EMBEDDED
Edit Copy Delete	24ITP31	Data Structures Lab	2	PRACTICAL
Edit Copy Delete	24ITR31	Data Structures	4	THEORY
Edit Copy Delete	24ITR32	Database Management Systems	4	THEORY
Edit Copy Delete	24ITR33	Artficial Intelligence	3	THEORY

ENROLLMENTS DATABASE:

	enrollment_id	roll_number	course_code	semester	total_classes	attended_classes
Edit Copy Delete	1	24ITR001	24ITR31	3	50	48
Edit Copy Delete	2	24ITR001	24ITR32	3	50	46
Edit Copy Delete	3	24ITR001	24ITP31	3	30	28
Edit Copy Delete	4	24ITR001	24ITE31	3	40	38
Edit Copy Delete	5	24ITR002	24ITR31	3	50	40
Edit Copy Delete	6	24ITR002	24ITR32	3	50	35
Edit Copy Delete	7	24ITR002	24ITP31	3	30	26
Edit Copy Delete	8	24ITR002	24ITE31	3	40	32
Edit Copy Delete	9	24ITR003	24ITR31	3	50	38
Edit Copy Delete	10	24ITR003	24ITR32	3	50	30
Edit Copy Delete	11	24ITR003	24ITP31	3	30	24
Edit Copy Delete	12	24ITR003	24ITE31	3	40	34
Edit Copy Delete	13	24ITR004	24ITR31	3	50	42
Edit Copy Delete	14	24ITR004	24ITR32	3	50	39
Edit Copy Delete	15	24ITR004	24ITP31	3	30	25
Edit Copy Delete	16	24ITR004	24ITE31	3	40	30
Edit Copy Delete	17	24ECE001	24ECR31	3	50	49
Edit Copy Delete	18	24ECE001	24ECR32	3	50	47
Edit Copy Delete	19	24ECE001	24ECP31	3	30	29
Edit Copy Delete	20	24ECE001	24ECE31	3	40	39
Edit Copy Delete	21	24ECE002	24ECR31	3	50	34
Edit Copy Delete	22	24ECE002	24ECR32	3	50	33
Edit Copy Delete	23	24ECE002	24ECP31	3	30	23
Edit Copy Delete	24	24ECE002	24ECE31	3	40	36
Edit Copy Delete	25	24ECE003	24ECR31	3	50	28

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

REFERENCES:

1. Oracle, Java SE Documentation – JDBC Overview and Architecture. https://docs.oracle.com/javase/8/docs/technotes/guides/jdbc/ .

2. Oracle, Java SE API – HttpServer. https://docs.oracle.com/en/java/javase/17/docs/api/jdk.httpserver/com/sun/net/httpserver/HttpServer.html .

3. Oracle, MySQL 8.0 Reference Manual. https://dev.mysql.com/doc/refman/8.0/en/ .

4. Oracle, MySQL Connector/J Developer Guide. https://dev.mysql.com/doc/connector-j/8.0/en/ .

5. University Grants Commission, Guidelines for Choice Based Credit System (CBCS). https://www.ugc.gov.in/pdfnews/8023719_Guidelines-for-CBCS.pdf .

6. Chart.js, Chart.js Documentation. https://www.chartjs.org/docs/latest/ .

7. Tailwind CSS, Tailwind CSS Documentation. https://tailwindcss.com/docs .

8. GeeksforGeeks, Establishing JDBC Connection in Java. https://www.geeksforgeeks.org/establishing-jdbc-connection-in-java/ .

9. YouTube, Java JDBC with MySQL Tutorial Videos. https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

10. Anthropic, Claude AI Assistant (used for project ideas and guidance). https://claude.ai/ .

GITHUB:

LINK: https://github.com/Srimanikandaprabhu-svg/Student-Academic-Continuous-Evaluation-System-using-Java-JDBC.git